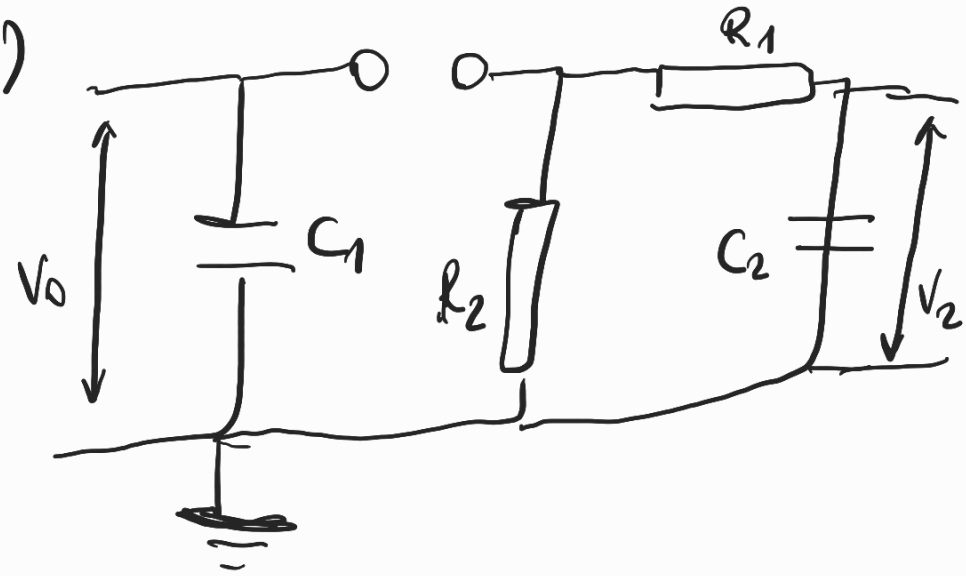
QUESTION-2
(25pts): Please,
a) Draw a single stage impulse generator circuit.

b) Explain **the role of each component** in the circuit.

c) Explain **how one impulse is generated** by the circuit, **step by step, in details**.

a)



b)

R_1 ⇒ Determines the charging speed, also charging time of C_2.

R_2 → Together with R_1, determines discharging speed.

C_1 and C_2 → Both of them should be fully charged for apply the experiment.

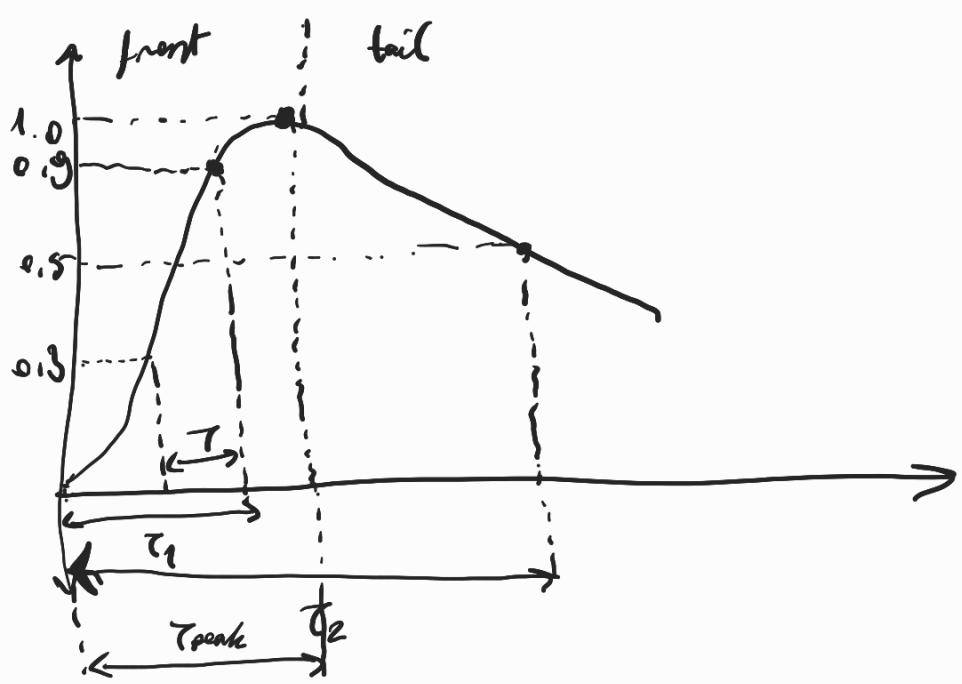
c) 1) We should set distance between couple spheres.

2) Charge C1 slowly.

3) When C1 reaches the optimum value, which is distance between spheres cannot resist, the arc will be happened current flowes.

4) Current flowes to C_2 and C_2 creates the front side of graph.

5) After maximum Voltage of impulse, C_2 starts discharging over $R_1.R_2$; and creates the tail side of curve.



QUESTION-3 (30 pts): In a planar electrode system, electrode gap is **K *mm*,** electrode surface area is **15 cm^2**, relative dielectric constant (relative permittivity) is **L**, applied voltage is **M× 30 kV,** dielectric breakdown strength of the insulation Material is **N× 15 kV/cm**.

Calculate: **(6 pts each)**

a) Electric field strength,

b) Capacity,

c) Charge,

d) Electric flux density,

e) The maximum voltage that can be applied.

Equations:

$$Q = C.U$$

$$D = \varepsilon.E$$

$$C = \frac{\varepsilon.S}{a}$$

$$E = \frac{U}{a}$$

$$\varepsilon_0 = 8{,}854.10^{-12}\ \text{F/m}$$

$$\varepsilon = \frac{U}{a} = \frac{60\ kV}{0{,}1\ cm} = 600\ \frac{kV}{cm}$$

b) $C = \frac{\varepsilon . S}{a} = \frac{\varepsilon . 15.10^{-4}}{1.10^{-3}}$

c) $Q = C.U = C.30.10^3$

QUESTION-4 (15 pts): Please choose and answer **ONLY ONE OF THE FOLLOWING**:

Option 1: Draw and explain in details how high voltage DC is generated using **electrostatic generator**.

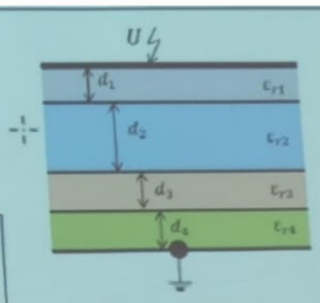
Option 2: Draw and explain in details how high voltage AC is generated using **two stage cascaded transformers.**

QUESTION-1 (35pts): In the multilayer planar electrode system with four dielectric layers shown in the figure below,

U=80 kV, $d_1 = 0{,}3$ cm, $d_2 = 0{,}7$ cm, $d_3 = 0{,}5$ cm, $d_4 = 0{,}6$ cm, $\varepsilon_{r1} = 1{,}3$, $\varepsilon_{r2} = 2{,}6$, $\varepsilon_{r3} = 0{,}9$, and $\varepsilon_{r4} = 0{,}5$.

a) (15 pts) Calculate the potential, minimum and maximum electric fields at each layer.

b) (20 pts) Considering breakdown strength for the layers as $E_{b1} = 105$ kV/cm, $E_{b2} = 80$ kV/cm, $E_{b3} = 35$ kV/cm and $E_{b4} = 48$ kV/cm find out if the system will withstand or not.



Equations:

$$A = \sum_{i=1}^{N} \frac{d_i}{\varepsilon_i} \qquad U_i = \frac{U\,d_i}{A\,\varepsilon_i} \qquad E_i = \frac{U}{A.\varepsilon_i} = \frac{U_i}{d_i}$$

$$U_i = \frac{U}{A} \cdot \frac{d_i}{\varepsilon_i}$$

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$$A = \frac{1}{\varepsilon_0}\left(\frac{0{,}3}{1{,}3} + \frac{0{,}7}{2{,}6} + \frac{0{,}5}{0{,}9} + \frac{0{,}6}{0{,}5}\right) = \frac{2{,}25}{\varepsilon_0}$$

$$U_i = \frac{U}{2{,}25} \cdot \frac{d_i}{\varepsilon_i}$$

$$U_1 = \frac{80\ kV}{2{,}25} \cdot \frac{0{,}3}{1{,}3} = 8{,}21\ kV$$

$$U_2 = 9{,}57\ kV$$

$$U_3 = 19{,}25\ kV$$

$$U_4 = 42{,}97\ kV$$

$$E_1 = \frac{U_1}{d_1} = \frac{8{,}21\ kV}{0{,}3\ cm} = 27{,}37\ \frac{kV}{cm}$$

$$E_2 = 13{,}67\ kV/cm$$

$$E_3 = 39{,}5\ kV/cm$$

$$E_4 =$$

QUESTION-2 (35 pts): In a planar electrode system, electrode gap is **6 *m***, electrode surface area is **18 *mm*²**, relative dielectric constant (relative permittivity) is 3, applied voltage is **22 kV** and dielectric breakdown strength of the insulation material is **79 kV/cm**. **Calculate**: (a) Electric field strength, (b) Capacity, Charge, (c) Electric flux density, (d) The maximum voltage that can be applied.

The formulas you may need:

$Q = C.U$ $\qquad$ $D = \varepsilon.E$

$C = C = \frac{\varepsilon.S}{a}$ $\qquad$ $E = \frac{U}{a}$

a) $\varepsilon = \frac{U}{a}$ → Main Voltage (kV); a → Elc. Gap (cm)

$\varepsilon = \frac{U}{a} = \frac{22\ kV}{600\ cm} = 0{,}036\ \frac{kV}{cm}$

b) $C = \frac{\varepsilon . S}{a} = \frac{8{,}85 . 10^{-12} . 3 . 18 . 10^{-6}}{6\ m} = 7{,}965 . 10^{-17}\ F$

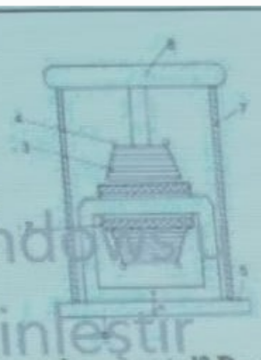
$Q = C . U = 7{,}965 . 10^{-17} . 22 . 10^{3} = 1{,}75 . 10^{-12}\ C$

c) $D = \varepsilon . E = 8{,}85 . 10^{-12} . 0{,}036 . 10^{3} . 10^{3}$

d) $V_{max} = \text{Breakdown Str.} \cdot \text{gap} = \frac{79\ kV}{cm} \cdot \frac{100\ cm}{1\ m} \cdot 6 = 4{,}74 . 10^{5} = 474\ kV$

QUESTION-3 (30 pts):

a) **(15 pts)** In the figure given below, a **high voltage AC testing transformer** is shown. Please write the **names of each numbered section** of the trafo and **briefly describe the role of each component**.



b) **(15 pts)** How **impulse voltages** can be generated? **Draw an** example impulse voltage generator and describe the role of each circuit component, and explain how each component can be selected/sized.

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